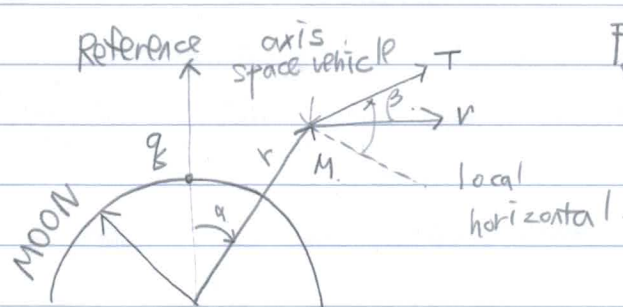


1) Example 5.1-2



$$F_g(t) = M g_0 R^2 / r^2(t)$$

Let $x_1 \triangleq r$, $x_2 \triangleq \alpha$, $x_3 \triangleq \dot{r}$, $x_4 \triangleq r\dot{\alpha}$ & $u \triangleq \beta$

$$\dot{x}_1(t) = x_3(t)$$

$$\dot{x}_2(t) = \frac{x_4(t)}{x_1(t)}$$

$$\dot{x}_3(t) = \frac{x_4^2(t)}{x_1(t)} - \frac{g_0 R^2}{x_1^2(t)} + \left[\frac{I}{M} \right] \sin u(t)$$

$$\dot{x}_4(t) = -\frac{x_3(t)x_4(t)}{x_1(t)} + \left[\frac{I}{M} \right] \cos u(t)$$

Mission a. Launch at point g ($t=0$). circular orbit of altitude D .
minimum time, $\alpha(t_f)$ is unspecified. IC: $\bar{x}(0) = [R \ 0 \ 0 \ 0]^T$

$$H = 1 + p_1(t)x_3(t) + \frac{p_2(t)x_4(t)}{x_1(t)} + p_3(t) \left[\frac{x_4^2(t)}{x_1(t)} - \frac{g_0 R^2}{x_1^2(t)} + \left[\frac{I}{M} \right] \sin u(t) \right] \\ + p_4(t) \left[-\frac{x_3(t)x_4(t)}{x_1(t)} + \left[\frac{I}{M} \right] \cos u(t) \right]$$

$$\begin{aligned} \dot{p}_1^*(t) &= -\frac{\partial H}{\partial x_1} = \frac{p_2^*(t)x_4^*(t)}{x_1^{*2}(t)} + p_3^*(t) \left[\frac{x_4^{*2}(t)}{x_1^{*2}(t)} - \frac{2g_0 R^2}{x_1^{*3}(t)} \right] - \frac{p_4^*(t)x_3^*(t)x_4^*(t)}{x_1^{*2}(t)} \\ \dot{p}_2^*(t) &= -\frac{\partial H}{\partial x_2} = 0 \\ \dot{p}_3^*(t) &= -\frac{\partial H}{\partial x_3} = -p_1^*(t) + \frac{p_4^*(t)x_4^*(t)}{x_1^*(t)} \\ \dot{p}_4^*(t) &= -\frac{\partial H}{\partial x_4} = -\frac{p_2^*(t)}{x_1^*(t)} - \frac{2p_3^*(t)x_4^*(t)}{x_1^*(t)} + \frac{p_4^*(t)x_3^*(t)}{x_1^*(t)} \end{aligned}$$

state equations: $\dot{\bar{x}}^*(t) = \bar{a}(\bar{x}^*(t), u^*(t))$

$$0 = \frac{\partial H}{\partial u} = \left[\frac{I}{M} \right] [p_3^*(t) \cos u^*(t) - p_4^*(t) \sin u^*(t)]$$

$$\rightarrow u^*(t) = \tan^{-1} \frac{p_3^*(t)}{p_4^*(t)}$$

$$\sin u^*(t) = \frac{p_3^*(t)}{\sqrt{p_3^{*2}(t) + p_4^{*2}(t)}}, \quad \cos u^*(t) = \frac{p_4^*(t)}{\sqrt{p_3^{*2}(t) + p_4^{*2}(t)}}$$

$$\text{B.C: } x_1^*(t_f) = R+D, \quad p_2^*(t_f) = 0$$

$$x_3^*(t_f) = 0, \quad x_4^*(t_f) = \sqrt{\frac{g_0 R^2}{R+D}}$$

$$H(\bar{x}^*(t_f), \bar{p}^*(t_f)) = 0$$

Mission b. Launch from point $Z(t=0)$ redevous (circular orbit D)

only B.C are changed: The final state (line on the moving point

$$\theta(t) = \begin{bmatrix} R+D \\ \text{modulo}(\pi t) \\ 0 \\ \pi(R+D) \end{bmatrix}$$

$$\delta \dot{x}_{2f} = \left[\frac{d\theta_2}{dt}(t_f) \right] \delta t_f = \pi \delta t_f$$

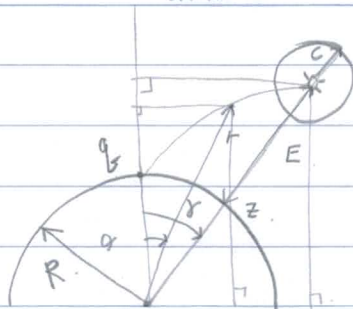
$$-\pi p_2^*(t_f) + H(\bar{x}^*(t_f), \bar{p}^*(t_f)) = 0$$

B.C :

$$\bar{x}^*(t_f) = \begin{bmatrix} R+D \\ \text{modulo}(\pi t_f) \\ 0 \\ \pi(R+D) \end{bmatrix} = \theta(t_f)$$

Mission c.

Synchronous orbit E miles above the point Z.
The space craft transmit television pictures at a distance of C miles



$$m(\bar{x}(t)) = [r(t) \cos \theta(t) - [R+E] \cos \delta]^2 + [r(t) \sin \theta(t) - [R+E] \sin \delta]^2 - C^2 = 0$$

$$H + \frac{\partial h}{\partial t} = \sum_{i=1}^k d_i \left[\frac{\partial m_i}{\partial t}(\bar{x}^*(t_f), t_f) \right] \quad (h=0)$$

$$\Rightarrow -\bar{p}^*(t_f) = d \left[\frac{\partial m}{\partial \bar{x}}(\bar{x}^*(t_f)) \right]$$

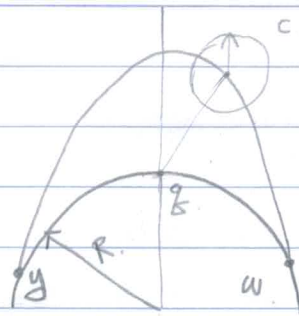
$$= 2d \begin{bmatrix} x_1^*(t_f) - [R+E] \cos(x_2^*(t_f) - \delta) \\ x_1^*(t_f) [R+E] \sin(x_2^*(t_f) - \delta) \\ 0 \\ 0 \end{bmatrix}$$

$$B.C: p_2^*(t_f) = 0, p_4^*(t_f) = 0$$

$$m(\bar{x}^*(t_f)) = [x_1^*(t_f) \cos x_2^*(t_f) - [R+E] \cos \delta]^2 + [x_1^*(t_f) \sin x_2^*(t_f) - [R+E] \sin \delta]^2 - C^2 = 0$$

$$H(\bar{x}^*(t_f), p^*(t_f)) = 0$$

Mission d.



spacecraft

From the radar data, the position history of the approaching

$$m(\vec{x}(t), t) = [r(t) \cos \theta(t) - 2.78 R t + 6.95 R t^2 - R]^2 + [r(t) \sin \theta(t) - 1.85 R t + 0.32 R]^2 - c^2 = 0.$$

$$-\vec{p}^*(t_f) = d \left[\frac{\partial m}{\partial \vec{x}}(\vec{x}^*(t_f), t_f) \right].$$

B.C:

$$-p_1^*(t_f) = 2d [x_1^*(t_f) + R \{ [-2.78 t_f + 6.95 t_f^2 - 1] \cos x_2^*(t_f) + [-1.85 t_f + 0.32] \sin x_2^*(t_f) \}]$$

$$-p_2^*(t_f) = -2d [R x_1^*(t_f) \{ [-2.78 t_f + 6.95 t_f^2 - 1] \sin x_2^*(t_f) + [1.85 t_f - 0.32] \cos x_2^*(t_f) \}]$$

$$-p_3^*(t_f) = 0.$$

$$-p_4^*(t_f) = 0.$$

$$[x_1^*(t_f) \cos x_2^*(t_f) - 2.78 R t_f + 6.95 R t_f^2 - R]^2 + [x_1^*(t_f) \sin x_2^*(t_f) - 1.85 R t_f + 0.32 R]^2 - c^2 = 0$$

$$H(z^*(t_f), \vec{p}^*(t_f)) = 2d R \{ [-2.78 + 13.9 t_f] [x_1^*(t_f) \cos x_2^*(t_f) - 2.78 R t_f + 6.95 R t_f^2 - R] - 1.85 [x_1^*(t_f) \sin x_2^*(t_f) - 1.85 R t_f + 0.32 R] \}$$

Example 5.2-1

Find the optimal control law for the system

$$\dot{x}(t) = ax(t) + u(t)$$

$$J(u) = \frac{1}{2} H x^2(T) + \int_0^T \frac{1}{4} u^2(t) dt$$

$$H = \frac{1}{4} u^2(t) + p^T(t) (ax(t) + u(t))$$

$$\dot{p}^*(t) = -\frac{\partial H}{\partial x} = -a p^*(t)$$

$$\begin{bmatrix} \dot{x}^*(t) \\ \dot{p}^*(t) \end{bmatrix} = \begin{bmatrix} a & -2 \\ 0 & -a \end{bmatrix} \begin{bmatrix} x^*(t) \\ p^*(t) \end{bmatrix}$$

$$\begin{aligned} \Phi(s) &= \begin{bmatrix} s-a & +2 \\ 0 & s+a \end{bmatrix}^{-1} = \frac{1}{(s-a)(s+a)} \begin{pmatrix} s+a & -2 \\ 0 & s-a \end{pmatrix} \\ &= \begin{bmatrix} \frac{1}{s-a} & \frac{-2}{(s-a)(s+a)} \\ 0 & \frac{1}{s+a} \end{bmatrix} = \frac{1}{a} \left(\frac{1}{s+a} - \frac{1}{s-a} \right) \end{aligned}$$

$$\Phi(t) = \begin{bmatrix} e^{at} & \frac{1}{a} e^{-at} - \frac{1}{a} e^{at} \\ 0 & e^{-at} \end{bmatrix}$$

$$K(t) = \begin{bmatrix} e^{-a(T-t)} & -\frac{H}{a} (e^{-a(T-t)} - e^{a(T-t)}) \end{bmatrix}^{-1} \begin{bmatrix} H & e^{a(T-t)} \end{bmatrix}$$

$$\therefore u^*(t) = -2K(t)x(t)$$

3). Find the N.C. to extremize the following performance index:

$$J = h_1(\vec{x}(t_0)) + h_2(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

t_0 is fixed & t_f is free.

$$\dot{\vec{x}} = f(\vec{x}(t), \vec{u}(t), t), \quad \vec{x} \in \mathbb{R}^n$$

$\vec{m}(\vec{x}(t_f), t_f) = \vec{0}$, $\vec{m} \in \mathbb{R}^k$, $k < n$: terminal conditions

$x_i(t_0) = \text{given}$, $x_i(t_f) = \text{free}$ $i = 1, 2, 3, \dots, k$

$x_j(t_0) = \text{free}$, $x_j(t_f) = \text{given}$ $j = k+1, k+2, \dots, n$

$$J_0 = h_1(\vec{x}(t_0), t_0) + h_2(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} [g + \vec{\lambda}^T (\vec{f} - \dot{\vec{x}})] dt \quad (\vec{\lambda}(t): \text{Lagrange multipliers})$$

$$= \left[\frac{\partial h_1}{\partial \vec{x}(t_0)} \delta \vec{x}(t_0) + \frac{\partial h_2}{\partial \vec{x}(t_f)} \delta \vec{x}(t_f) - \vec{\lambda}^T \delta \vec{x} \right]_{t_0}^{t_f}$$

$$+ \int_{t_0}^{t_f} \left[\left(\frac{\partial g}{\partial \vec{x}} + \vec{\lambda}^T \frac{\partial \vec{f}}{\partial \vec{x}} + \dot{\vec{\lambda}}^T \right) \delta \vec{x} + \left(\frac{\partial g}{\partial \vec{u}} + \vec{\lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} \right) \delta \vec{u} + (\vec{f} - \dot{\vec{x}})^T \delta \vec{\lambda} \right] dt = 0$$

$$\text{Let } H = g + \vec{\lambda}^T \vec{f}$$

when

$i = 1, 2, \dots, k$

$(\vec{x} = (x_1, x_2, \dots, x_k)^T)$

$$\frac{\partial H}{\partial \vec{u}} = 0, \quad \dot{\vec{\lambda}} = -\frac{\partial H}{\partial \vec{x}}, \quad \vec{x} = \frac{\partial H}{\partial \vec{\lambda}} = \vec{f}$$

$$\therefore \vec{x}^*(t_0) = \vec{x}_0$$

$$\frac{\partial h_2}{\partial \vec{x}}(\vec{x}^*(t_f), t_f) - \vec{\lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \vec{x}}(\vec{x}^*(t_f), t_f) \right]$$

$$\vec{m}(\vec{x}(t_f), t_f) = \vec{0}$$

$$H(\vec{x}^*(t_f), \vec{u}^*(t_f), \vec{\lambda}^*(t_f)) + \frac{\partial h_2}{\partial \vec{x}}(\vec{x}^*(t_f), t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \vec{x}}(\vec{x}^*(t_f), t_f) \right]$$

when

$j = k+1, k+2, \dots, n$

$(\vec{x} = (x_{k+1}, \dots, x_n)^T)$

$$\vec{x}^*(t_f) = \vec{x}_f$$

$$H(\vec{x}^*(t_f), \vec{u}^*(t_f), \vec{\lambda}^*(t_f), t_f) + \frac{\partial h_2}{\partial \vec{x}}(\vec{x}^*(t_f), t_f) = 0$$

$$(\int \vec{x}(t_0) \neq 0)$$

$$\frac{\partial h_1}{\partial \vec{x}}(\vec{x}^*(t_0), t_0) + \vec{\lambda}^*(t_0) = 0$$